L3 Body Composition

What the app does

The app takes one abdominal CT scan, finds the third lumbar vertebra (L3) in the lower back, and measures muscle and fat on that one slice. L3 is the standard level: its muscle and fat track the whole body's, and all published cut-offs use it.

Two AI models do the drawing (TotalSegmentator finds L3 and both psoas; a Comp2Comp model marks muscle and fat), then our script measures and `report.py` makes the colour report.

The tissues we measure

Tissue	Colour in report	What it is	Why it matters
Right psoas	Cyan	Long muscle beside the spine on the patient's right, joins the spine to the thigh; used to bend the hip	Easy to find and measure; a quick marker of overall muscle
Left psoas	Yellow	Same muscle on the patient's left	Its density (HU) is the key number in the mCRPC paper: low density = shorter survival
Other muscle	Red	All other skeletal muscle on the slice: back muscles beside the spine, belly-wall muscles (front and sides)	Together with both psoas it makes total muscle (SMA)
SAT — subcutaneous fat	Orange	Fat directly under the skin, the outer ring	Stored energy; part of overall body fat
VAT — visceral fat	Green	Fat inside the belly, around the intestines and organs	Linked to heart and metabolic risk; high VAT = visceral obesity
IMAT — intermuscular fat	Purple	Small streaks of fat between and inside the muscles	More IMAT = muscle being replaced by fat; an extra we report, not in the requirements

Two words come up often:

- **Sarcopenia** = too little muscle. Measured by SMI or PMI (below).
- **Myosteatosis** = fat inside the muscle, so the muscle looks darker on CT. Measured by the muscle's average HU (below).

HU: how dense each pixel is

Each CT pixel has a density number in HU: air −1000, fat −190 to −30, water 0, muscle −29 to +150 (healthy 40–60), bone +400 and up. The app counts only pixels in the right range for each tissue, and uses the muscle's average HU as a quality measure: **lower HU = more fat inside the muscle**.

The measurements

Areas = pixel count × pixel size. Indices = area ÷ height², so tall and short people can be compared. Examples are from the report image below (test phantom, height 1.70 m, so height² = 2.89).

Measurement	What it means	How it is calculated	Example
SMA	All muscle on the slice	Pixel count × pixel size	99.5 cm²
SMI	Muscle for your height	SMA ÷ height²	99.5 ÷ 2.89 = 34.4
Psoas area (R / L)	Size of each psoas	Pixel count × pixel size	5.16 / 5.46 cm²
A × B, C × D	Longest width of each psoas and the width across it	Measured on the slice (the cross lines)	32.8 × 21.6 mm, 34.4 × 21.6 mm
PMI (diameters)	Psoas for your height, from widths	(A·B + C·D) ÷ height², in cm	(3.28·2.16 + 3.44·2.16) ÷ 2.89 = 5.02
PMI (area)	Psoas for your height, from areas	(R + L area) ÷ height²	(5.16 + 5.46) ÷ 2.89 = 3.68
SAT / SATI	Fat under the skin	Area; area ÷ height²	128.9 cm² / 44.6
VAT / VATI	Fat inside the belly	Area; area ÷ height²	107.7 cm² / 37.3
IMAT	Fat between muscles	Area	2.10 cm²
Muscle HU	Muscle quality	Average HU of all muscle	62.8 HU
Left / right psoas HU	Psoas quality	Average HU of that psoas	52.3 / 49.5 HU

Without height, all the indices stay blank. Weight is optional and only gives BMI for the muscle HU cut-off.

Units in the report. Areas (SMA, psoas, SAT, VAT, IMAT) are in **cm²**. Psoas widths A, B, C, D are in **mm**. Indices (SMI, PMI, SATI, VATI) are area ÷ height², in **cm²/m²**. Density is in **HU**. For PMI the widths are first changed from mm to cm (÷ 10): 32.8 mm = 3.28 cm. Compare an index with an index cut-off, and an area with an area cut-off: SMI 34.4 cm²/m² against 52.4 cm²/m², VAT 107.7 cm² against 100 cm².

Normal values and cut-offs

Each measure has a published cut-off that splits normal from low or high. The app flags five of them today; the others are given for reference.

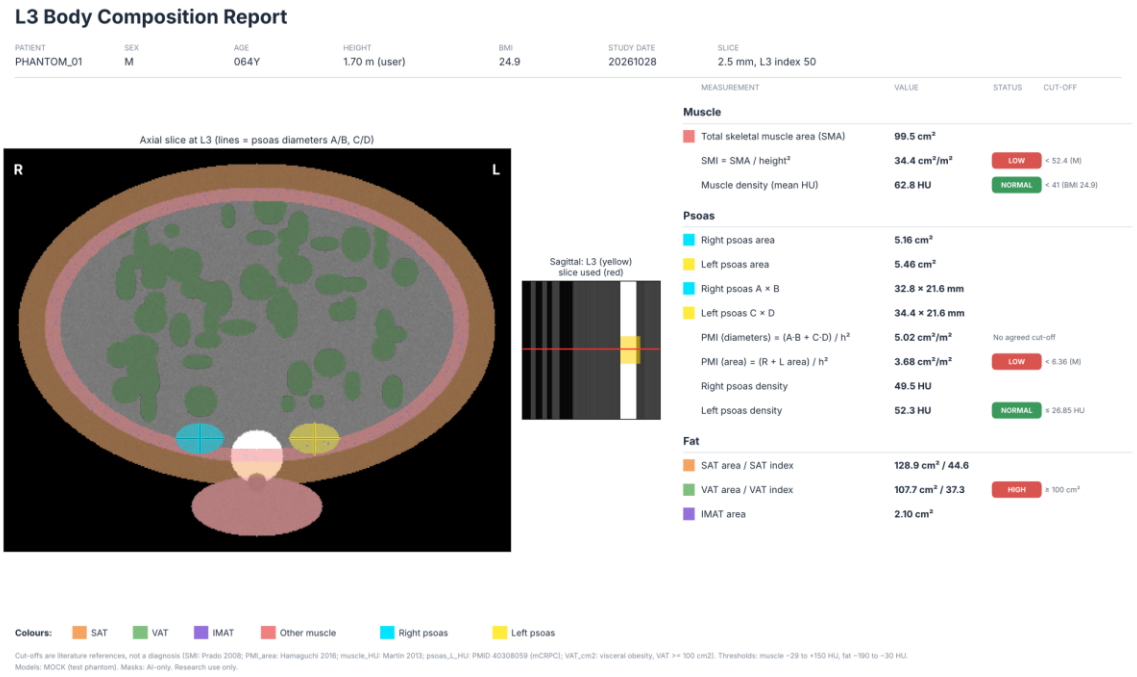
Measurement	Normal (healthy side)	Low or high = risk	Source	Flag in app
SMI (cm^2/m^2)	Men ≥ 52.4, women ≥ 38.5	**Low** below that = too little muscle (sarcopenia). Martin uses men < 43 (BMI < 25) or < 53 (BMI ≥ 25), women < 41	Prado 2008; Martin 2013	Yes (Prado)
PMI area (cm^2/m^2)	Men ≥ 6.36, women ≥ 3.92	**Low** below that = small psoas	Hamaguchi 2016 (healthy Japanese adults)	Yes
PMI diameters (cm^2/m^2)	No agreed value	Higher = more psoas; compare patients or time points	—	No
Muscle HU	≥ 41 HU if BMI < 25; ≥ 33 HU if BMI ≥ 25	**Low** below that = fat inside muscle (myosteatosis)	Martin 2013	Yes
Left psoas HU	> 26.85 HU	**Low** at or below 26.85 = shorter survival in mCRPC before Lu-177	PMID 40308059	Yes
VAT area (cm^2)	< 100	**High** at or above 100 = visceral obesity, more heart and metabolic risk	Common definition, mainly Asian studies	Yes
VATI (cm^2/m^2)	Men < 52.9, women < 51.5	**High** above that = high visceral fat	Ebadi 2017 (cancer patients)	No
SAT area / SATI (cm^2/m^2)	SATI men ≥ 50, women ≥ 42 linked to better survival in cancer	**Low** SAT = fewer energy reserves; in cancer, low is the worse side	Ebadi 2017 (cancer patients)	No
IMAT (cm^2)	No agreed value; lower is better	Higher = more fat between muscles	—	No

For muscle, there is no "too high": above the cut-off is simply normal. For fat, VAT is the one where high is bad; in cancer patients, low SAT is the warning sign. The values come from different groups of patients, so they are a guide, not a diagnosis.

A grey flag (NEEDS SEX, NEEDS BMI, NO HEIGHT) means information is missing, not that something is wrong.

How to read the report

The report is one page with four parts. This example was made from a test phantom (a fake body), not a real patient.



1. **Header:** patient, sex, age, height and its source, BMI, slice used. Check height first.
2. **Left image:** the L3 slice with coloured tissues. Patient's right is on the screen's left; cross lines = psoas widths.
3. **Middle image:** side view; L3 in yellow, the red line = the slice used.
4. **Right table:** every value with its flag and cut-off.

Quick check: does the report look right?

- ☐ The red line crosses the middle of the yellow vertebra (L3).
- ☐ Cyan and yellow sit as two round shapes either side of the spine.
- ☐ Orange is a ring under the skin; green is only inside the belly.
- ☐ No colour sits on bone (white) or gas (black).
- ☐ Height is real, not missing or a placeholder.
- ☐ Values look believable: SMA 80–200 cm², muscle HU 20–60, each psoas 3–15 cm².

If the app says "L3 vertebra not found", the scan does not reach L3.